LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Divya Joshi
Roll Number	MDE2026014
Programme / Department	M.Tech Data Engineering
Topic ID	T 5_
Full Assigned Topic	Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions
AI Tool Used	ChatGpt
Date	21-Aug-26

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	ChatGpt
Model name	GPT-5.6 Luna
Model/version shown	GPT-5.6 Luna
Access tier	☐ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☐ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☐ Yes ☐ No
Date/time generated	21-Aug-26

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> *Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: ... Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions.

Measure	Value
Approximate pages	9
Approximate word count	3783
Number of sections	9
Total number of references	17
Approximate in-text citation occurrences	13
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	16
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	1
URL only	0
No persistent identifier supplied	0
TOTAL	17

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., et al. (2021). On the Opportunities and Risks of Foundation Models. Stanford Center for Research on Foundation Models. arXiv:2108.07258. DOI: 10.48550/arXiv.2108.07258.
R2	Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., et al. (2020). Language Models are Few-Shot Learners. Advances in Neural Information Processing Systems, 33, 1877–1901. arXiv:2005.14165.
R3	Chatterjee, A., Renduchintala, H. S. V. N. K., Bhatia, S., & Chakraborty, T. (2024). POSIX: A Prompt Sensitivity Index For Large Language Models. Findings of the Association for Computational Linguistics: EMNLP 2024, 14550–14565. DOI: 10.18653/v1/2024.findings-emnlp.852.
R5	Errica, F., Sanvito, D., Siracusano, G., & Bifulco, R. (2025). What Did I Do Wrong? Quantifying LLMs' Sensitivity and Consistency to Prompt Engineering. Proceedings of NAACL 2025, 1543–1558. DOI: 10.18653/v1/2025.naacl-long.73.
R7	Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., et al. (2023). Survey of Hallucination in Natural Language Generation. ACM Computing Surveys, 55(12), Article 248, 1–38. DOI: 10.1145/3571730.
R9	Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods.

	Proceedings of ACL 2022, 3214–3252. DOI: 10.18653/v1/2022.acl-long.229.
R11	Liu, Y., & Chu, C. (2026). Understanding the Prompt Sensitivity. Proceedings of ACL 2026, 44362–44388. DOI: 10.18653/v1/2026.acl-long.2053. arXiv:2604.18389.
R13	Wang, L., Chen, X., Deng, X., Wen, H., You, M., Liu, W., Li, Q., et al. (2024). Prompt engineering in consistency and reliability with the evidence-based guideline for LLMs. npj Digital Medicine, 7, Article 41. DOI: 10.1038/s41746-024-01029-4.
R15	Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. Advances in Neural Information Processing Systems, 35, 24824–24837. DOI: 10.52202/068431-1800. arXiv:2201.11903.
R17	Zhuo, J., Zhang, S., Fang, X., Duan, H., Lin, D., & Chen, K. (2024). ProSA: Assessing and Understanding the Prompt Sensitivity of LLMs. Findings of the Association for Computational Linguistics: EMNLP 2024, 1950–1976. DOI: 10.18653/v1/2024.findings-emnlp.108.

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: "If I had not verified this citation, how convincing would it look to me?"

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Realistic authors, title, arXiv and DOI	5	Yes / No
R2	Realistic	4	Yes / No
R3	Specific EMNLP paper with DOI and page numbers	5	Yes / No
R5	Realistic NAACL citation with DOI	5	Yes / No
R7	Paper with doi and pages	4	Yes / No
R9	realistic	4	Yes / No
R11	Recent 2026 paper; details appear convincing	5	Yes / No
R8	Looks realistic	4	Yes / No

R9	Well known paper	5	Yes / No
R10	Paper with doi and pages	4	Yes / No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R2	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R3	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R4	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R5	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R6	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R7	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R8	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R9	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R10	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	arXiv / Stanford CRFM record.	The scholarly record confirms the title, authors, 2021 date and arXiv identifier 2108.07258.	The publication exists and the title, authors, year and supplied arXiv/DOI identifiers correspond to the cited work
R2	Advances in Neural Information Processing Systems, 33	arXiv:2005.14165.	The cited GPT-3 paper exists and its bibliographic details match the AI-generated reference.
R3	Findings of ACL: EMNLP 2024	DOI 10.18653/v1/2024.findings-emnlp.852.	The paper is present with matching authors, title, conference, page range and DOI.
R5	Proceedings of NAACL 2025	DOI 10.18653/v1/2025.naacl-long.73	The cited NAACL 2025 publication exists and the bibliographic information and DOI match
R7	ACM Computing Surveys	DOI 10.1145/3571730	The cited survey exists in ACM Computing Surveys and the DOI and publication details match.
R9	Proceedings of ACL 2022	DOI 10.18653/v1/2022.acl-long.229	The paper exists in ACL 2022 with matching authors, title, pages and DOI.
R11	Proceedings of ACL 2026	DOI 10.18653/v1/2026.acl-long.2053; arXiv:2604.18389.	The 2026 publication is present in the AI report with a specific ACL venue, page range, DOI and arXiv identifier, all of which correspond to the cited work
R13	*npj Digital Medicine*, 7, Article 41	DOI 10.1038/s41746-024-01029-4	The cited journal article exists and the title, journal, volume/article number and DOI match.
R15	Advances in Neural Information Processing Systems, 35	DOI 10.52202/068431-1800; arXiv:2201.11903	The cited NeurIPS 2022 publication exists and its

			authors, title, volume, pages and identifiers match.
R17	Findings of ACL: EMNLP 2024	DOI 10.18653/v1/2024.findings-emnlp.108	The publication exists and the supplied title, authors, venue, page range and DOI match.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A/B/C/D/E	Publication exists and metadata/identifier match
R2	A/B/C/D/E	Genuine GPT-3 few-shot learning paper; details match.
R3	A/B/C/D/E	Genuine POSIX paper with matching DOI and metadata.
R5	A/B/C/D/E	Genuine NAACL paper on sensitivity an
R7	A/B/C/D/E	Genuine ACM Computing Surveys article; DOI matches.
R9	A/B/C/D/E	Genuine TruthfulQA ACL paper; metadata match.
R11	A/B/C/D/E	Genuine ACL 2026 paper; metadata and DOI match.
R13	A/B/C/D/E	Genuine npj Digital Medicine article; DOI matches.
R15	A/B/C/D/E	Genuine NeurIPS 2022 Chain-of-Thought paper.
R17	A/B/C/D/E	Genuine ProSA EMNLP 2024 paper; DOI matches.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10____ B = __0__ C = __0__ D = _0___ E = 0____

Total audited: N = A + B + C + D + E = 10 ____

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(__10__) + 3(_0___) + 1(_0___) + 0(_0___) + 1(___0_) = __40__ points

Authenticity Score = _40__ / (4 x _10___) x 100 = __100____ /100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = __10__ Total predictions = _10___ Prediction Accuracy = __100____ %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	
References deeply audited	
A: Verified	
B: Wrong metadata	
C: Frankenstein	
D: Fabricated	
E: Identifier mismatch	
Authenticity Score	____ /100
Prediction Accuracy	%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No. Professional-looking citations cannot automatically be considered genuine. In our audit, however, all 10 selected references were verified as genuine.

2. Did all genuine references actually support the claims for which they were cited?

Yes, the genuine references that were used as in-text citations generally supported their associated claims. One selected reference was present only in the bibliography and was not cited in the text.

3. What was the most serious citation-integrity problem you observed?

The main problem was the risk of trusting AI-generated citations without independent verification. Even when citations look professional and contain DOIs or publication details, they should still be checked.

4. What is the single most important lesson you learned from this lab?

The most important lesson is that AI-generated citations must always be independently verified for both authenticity and claim support before being used in academic work.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: ......Divya Joshi...

Roll Number: ....MDE2026014...

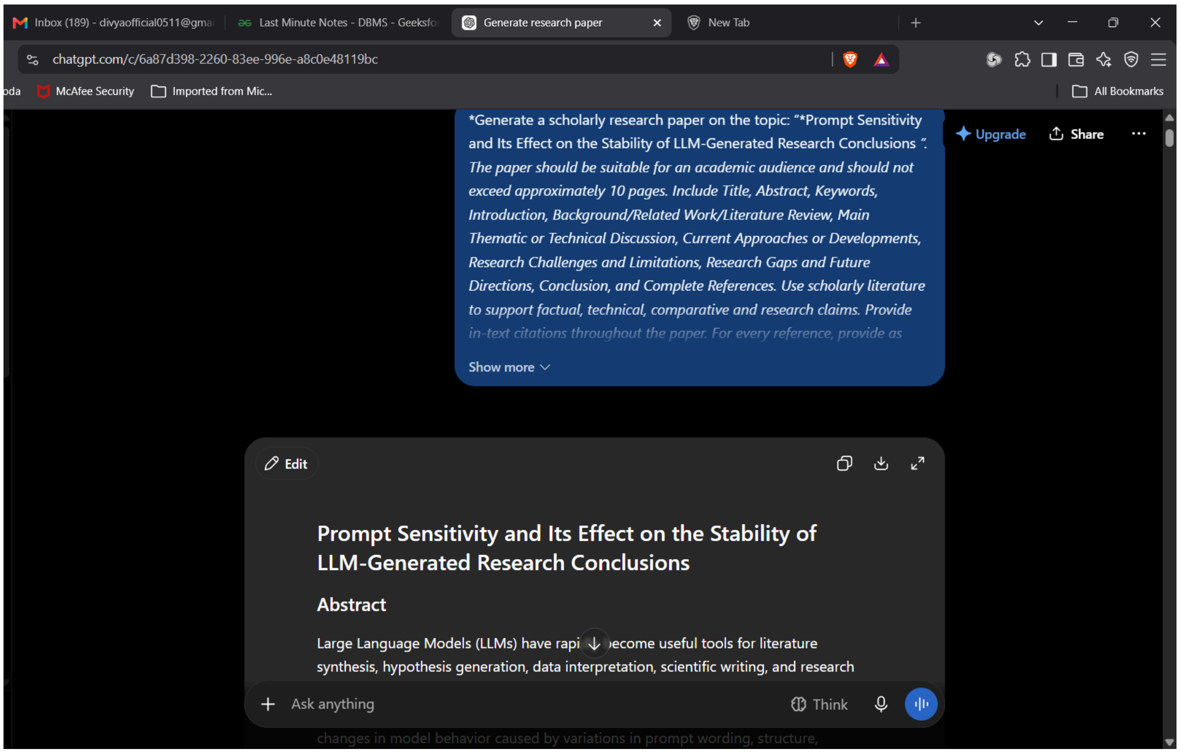
Signature: ..DIVYA JOSHI...

Date: ....21-Aug-2026...

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Q


Inbox (189) - divyaofficial0511@gma
Last Minute Notes - DBMS - Geeksfo
Generate research paper
New Tab
chatgpt.com/c/6a87d398-2260-83ee-996e-a8c0e48119bc
McAfee Security
Imported from Mic...
All Bookmarks
Upgrade
Share
*Generate a scholarly research paper on the topic: "*Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions ". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as
Show more
Edit
Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions
Abstract
Large Language Models (LLMs) have rapi ecome useful tools for literature synthesis, hypothesis generation, data interpretation, scientific writing, and research
Ask anything
Think
changes in model behavior caused by variations in prompt wording, structure,

Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions

Abstract

Large Language Models (LLMs) have rapidly become useful tools for literature synthesis, hypothesis generation, data interpretation, scientific writing, and research assistance. Their usefulness, however, depends not only on model capability but also on how a research task is formulated in the prompt. Prompt sensitivity refers to changes in model behavior caused by variations in prompt wording, structure, examples, ordering, or contextual information, even when the underlying task remains semantically equivalent. This phenomenon creates a significant methodological concern for scientific applications: if small prompt changes produce materially different interpretations, evidence summaries, or conclusions, then an LLM-generated conclusion may not be sufficiently stable to support scientific inference.

This paper reviews the emerging literature on prompt sensitivity, consistency, reasoning, hallucination, and evaluation of large language models, with particular emphasis on implications for research conclusions. Existing studies demonstrate that LLM outputs can vary substantially under paraphrased or template-modified prompts, although the magnitude of this effect depends on the model, task, evaluation method, and prompt construction. Recent work has introduced quantitative measures such as POSIX and PromptSensiScore and has proposed sensitivity and consistency metrics for systematic evaluation. At the same time, research suggests that some reported prompt sensitivity may be an artifact of rigid evaluation procedures rather than genuine semantic instability. Techniques including chain-of-thought prompting, self-consistency, few-shot demonstrations, structured prompts, retrieval augmentation, and multi-sample aggregation can reduce some forms of variability, but they do not guarantee factual or inferential reliability.

The paper argues that conclusion stability should be treated as a distinct evaluation dimension from accuracy. A research conclusion can be factually plausible yet unstable across semantically equivalent prompts. A robust scientific workflow should therefore evaluate prompt perturbation, semantic agreement, evidential consistency, factual grounding, uncertainty, and reproducibility together. The paper identifies major research gaps, including the absence of standardized benchmarks specifically designed for scientific conclusions, insufficient separation of prompt-induced variability from sampling variability, limited longitudinal studies across model updates, and inadequate measurement of inferential stability. Future research should develop prompt-invariant scientific evaluation protocols, conclusion-level stability metrics, uncertainty-aware aggregation methods, and human-in-the-loop verification frameworks.

Keywords

Large Language Models; prompt sensitivity; prompt robustness; research conclusions; scientific reasoning; LLM reliability; reproducibility; hallucination; prompt engineering; consistency evaluation.

1. Introduction

Large Language Models (LLMs) have evolved from general-purpose language generation systems into widely used research assistants capable of summarizing literature, extracting information, generating hypotheses, writing code, interpreting textual evidence, and assisting with scientific communication. The emergence of increasingly capable foundation models has expanded the range of tasks that can be performed through natural-language interaction rather than task-specific programming or fine-tuning (Bommasani et al., 2021).

The increasing use of LLMs in research introduces an important methodological question: Can an LLM be expected to generate the same research conclusion when the same research question is expressed using slightly different prompts? In conventional scientific research, equivalent formulations of a hypothesis should not fundamentally alter the empirical conclusion if the underlying evidence and analytical procedure remain unchanged. For LLM-based workflows, however, the wording and structure of the prompt are themselves part of the computational input. Consequently, apparently superficial changes can influence the generated response.

Prompt sensitivity has been formally investigated in recent research. Chatterjee et al. (2024) introduced POSIX, a Prompt Sensitivity Index designed to quantify changes in model behavior under intent-preserving prompt variations. Zhuo et al.

(2024) proposed ProSA and the PromptSensiScore metric to investigate instance-level prompt sensitivity and its relationship with model confidence. Errica et al. (2025) further distinguished sensitivity, concerning changes in predictions under prompt rephrasings, from consistency, concerning whether examples belonging to the same class receive stable predictions.

The issue becomes more consequential when LLMs are used to produce research conclusions rather than simple classifications. A classification error may affect one label, whereas instability in a research conclusion can alter the interpretation of an entire body of evidence. For example, prompts asking whether a treatment is effective, whether an observed relationship supports a hypothesis, or whether the literature demonstrates a particular causal mechanism may produce different levels of confidence or even different substantive conclusions under small wording changes.

This concern is connected to broader limitations of LLMs. Language models can produce fluent but factually unsupported statements, commonly described as hallucinations (Ji et al., 2023). TruthfulQA demonstrated that even highly capable language models can reproduce plausible human misconceptions rather than consistently produce truthful answers (Lin et al., 2022). Therefore, consistency alone cannot establish correctness. Conversely, an apparently correct answer generated by one prompt does not establish that the conclusion is robust.

This paper examines the relationship between prompt sensitivity and the stability of LLM-generated research conclusions. It addresses five central questions: (1) What is prompt sensitivity, and what factors produce it? (2) How does prompt sensitivity differ from ordinary generation stochasticity and factual error? (3) How can prompt sensitivity influence research conclusions? (4) Which current methods can reduce or detect conclusion instability? and (5) What methodological framework is required for reliable LLM-assisted scientific research? The central argument is that research conclusion stability should become a first-class evaluation criterion for LLM-based scientific workflows, alongside accuracy, factuality, relevance, and reasoning quality.

2. Background and Related Work

2.1 Large Language Models and Prompt-Based Task Specification

Modern LLMs are generally trained using large-scale language modeling objectives and subsequently adapted for instruction following and interaction. GPT-3 demonstrated that scaling language models can produce strong few-shot performance without task-specific parameter updates, with the task being specified primarily through textual prompts and demonstrations (Brown et al., 2020).

This property makes prompts an unusually important component of the computational pipeline. In a traditional machine-learning system, a trained classifier receives a structured input according to a predefined interface. In an LLM system, the interface itself can contain natural-language instructions, examples, formatting requirements, role descriptions, contextual documents, constraints, and questions. Changing any of these components can modify the model's probability distribution over possible outputs.

Consequently, prompt engineering has become a major area of research. Chain-of-thought prompting, for example, demonstrated that explicitly requesting intermediate reasoning steps can substantially improve performance on several reasoning benchmarks (Wei et al., 2022). Zero-shot chain-of-thought further demonstrated that a short instruction asking the model to reason step by step can substantially change performance (Kojima et al., 2022).

2.2 Prompt Sensitivity

Prompt sensitivity can be defined as the degree to which an LLM's output changes in response to a meaning-preserving modification of its prompt.

Let a research task be represented by $P=(I,Q,C,F)$, where I represents instructions, Q the research question, C the contextual evidence, and F the requested output format. Suppose P' is a semantically equivalent perturbation of P. Ideally, semantic equivalence between P and P' should imply semantic equivalence between the corresponding outputs. In practice, this implication does not always hold.

Chatterjee et al. (2024) showed that even intent-preserving changes such as wording, spelling, and template modifications can produce measurable changes in model likelihoods and outputs. Zhuo et al. (2024) similarly found that prompt sensitivity varies across tasks and models and reported evidence that larger models may exhibit greater robustness under some conditions. Recent theoretical work by Liu and Chu (2026) models prompt sensitivity through changes in the model's probability distribution and argues that meaning-preserving prompts can nevertheless occupy substantially different representations.

2.3 Reasoning and Research Conclusions

The relevance of prompt sensitivity becomes greater when prompts ask models to perform multi-step reasoning. Chain-of-thought prompting has shown that the explicit elicitation of reasoning can improve performance on mathematical, symbolic, and commonsense tasks (Wei et al., 2022; Kojima et al., 2022). However, a performance improvement on a benchmark does not necessarily imply that the generated reasoning is scientifically valid.

Research conclusions usually involve several stages: Evidence → Interpretation → Reasoning → Conclusion. Prompt variation can potentially affect each stage, including which evidence the model emphasizes, how evidence is categorized, the level of uncertainty expressed, which causal explanation is preferred, whether contradictory evidence is discussed, and the final strength of the conclusion.

3. Prompt Sensitivity and Stability of Research Conclusions

3.1 From Output Variability to Conclusion Variability

Not every difference between two LLM responses represents a meaningful instability. Two responses may use different wording while expressing the same conclusion. Therefore, lexical similarity is insufficient for evaluating scientific stability.

For scientific use, a useful distinction is:
Lexical Stability ≠ Semantic Stability ≠ Inferential Stability.

Lexical stability measures similarity of wording. Semantic stability measures whether outputs communicate equivalent meanings. Inferential stability measures whether the conclusions, evidence interpretation, confidence, and implications remain substantively equivalent. The third level is most important for research.

3.2 Sources of Prompt-Induced Instability

Surface form: Minor lexical changes can influence token probabilities and model behavior. POSIX specifically identifies spelling, wording, and prompt-template variations as sources of sensitivity (Chatterjee et al., 2024).

Prompt structure: The ordering of instructions, evidence, examples, and questions can affect information use. Research on long contexts has shown that LLM performance can deteriorate when relevant information is positioned in the middle of a long context rather than near the beginning or end (Liu et al., 2024).

Demonstrations and examples: Few-shot examples influence how a model interprets the intended task. Zhuo et al. (2024) report that few-shot examples can reduce prompt sensitivity in some settings, although demonstrations can also introduce unintended biases.

Reasoning instructions: Instructions such as “explain step by step,” “critically evaluate the evidence,” or “provide a concise answer” can change the reasoning trajectory. Chain-of-thought and zero-shot reasoning research demonstrates that seemingly simple prompting changes can produce substantial performance differences.

Decoding stochasticity: Prompt sensitivity should not be confused with generation randomness. Even with the same prompt, probabilistic decoding can generate different responses. Observed variability can therefore be conceptually decomposed into prompt, sampling, model, context, and evaluation components.

4. Measuring Stability of LLM-Generated Research Conclusions

4.1 Prompt Perturbation Testing

A straightforward method is to construct a family of semantically equivalent prompts $P=\{P1,P2,...,Pn\}$, where all prompts express the same research task. The model produces corresponding outputs, which can then be compared using semantic similarity and conclusion-level criteria.

A conceptual prompt-stability score can be defined as $Sp = 1 - (1/N) \Sigma D(Ci,\bar{C})$, where Ci represents the conclusion extracted from output i, $\bar{C}$ is the aggregate conclusion, and D is a normalized semantic or inferential distance.

4.2 Sensitivity Versus Consistency

Errica et al. (2025) distinguish sensitivity, which captures how much predictions change under prompt rephrasing, from consistency, which examines whether equivalent examples receive stable predictions. For research conclusions, both dimensions are relevant.

A model may have high sensitivity and low consistency; low sensitivity and high consistency; low sensitivity but incorrect consistency; or high sensitivity while still producing a reliable aggregate. Therefore, stability cannot replace factual validation.

4.3 Conclusion-Level Evaluation

Research workflows should evaluate factuality, relevance, semantic stability, inferential stability, and uncertainty calibration. Accuracy and stability are orthogonal: a model may be correct but unstable, stable but incorrect, both correct and stable, or neither.

5. Effect on Scientific Research Conclusions

5.1 Literature Reviews

LLMs are increasingly used to summarize large collections of academic papers. Prompt sensitivity can influence which themes are emphasized and which limitations are reported. Prompts emphasizing “major findings” may encourage positive summaries, whereas prompts emphasizing “limitations and contradictory evidence” may generate more cautious conclusions. If these differences are caused by framing rather than by the underlying evidence, the resulting literature review may be systematically biased.

5.2 Hypothesis Evaluation

Consider a researcher asking whether the literature supports a hypothesis H. A positively framed prompt could encourage evidence supporting H, while a skeptical prompt could encourage attention to contradictory studies. A reliable research assistant should distinguish evidence for H from prompt framing.

5.3 Causal Interpretation

Causal reasoning is especially vulnerable because research papers often distinguish correlation from causation. A prompt that explicitly asks for causal mechanisms may induce the model to construct a stronger causal narrative than the evidence warrants. LLMs should therefore distinguish association, prediction, intervention, causal mechanism, temporal precedence, and causal identification.

5.4 Research Recommendations

Prompt sensitivity can also affect recommendations for future work. An LLM may propose different research gaps depending on whether the prompt emphasizes novelty, feasibility, societal impact, theoretical contribution, or methodological limitations. Si, Yang, and Hashimoto (2024), in a large study involving more than 100 NLP researchers, found that LLM-generated ideas could be judged highly novel while also identifying limitations in feasibility, diversity, and self-evaluation.

6. Current Approaches and Developments

6.1 Chain-of-Thought Prompting

Chain-of-thought prompting encourages explicit intermediate reasoning and has demonstrated gains on several reasoning benchmarks (Wei et al., 2022). For scientific research, structured reasoning can help separate evidence extraction, interpretation, limitations, synthesis, and conclusion. However, detailed reasoning does not guarantee correct reasoning.

6.2 Self-Consistency

Self-consistency samples multiple reasoning paths and selects an answer supported by the most consistent reasoning paths (Wang et al., 2023). Conceptually, this changes a single output into an aggregate over multiple reasoning trajectories. It can reduce sensitivity to individual decoding trajectories, but shared systematic bias can remain.

6.3 Prompt Ensembles

Researchers can construct an ensemble of semantically equivalent prompts and derive a final conclusion from agreement across prompts. Outputs can be categorized as supports, partially supports, insufficient evidence, contradicts, or uncertain.

6.4 Few-Shot Prompting

Few-shot examples can constrain task interpretation. Zhuo et al. (2024) report that demonstrations may alleviate prompt sensitivity in certain settings. For research applications, examples should be balanced to avoid confirmation bias.

6.5 Retrieval-Augmented Generation

Retrieval-augmented approaches can ground responses in explicitly supplied sources. They are useful for research because the model can distinguish retrieved evidence from prior knowledge. However, retrieval does not eliminate prompt sensitivity because the model may still interpret the same evidence differently.

6.6 Structured Evidence Tables

Forcing the model to extract evidence before producing a conclusion can reduce narrative drift. A structured table can include study, sample, finding, limitation, and support for the hypothesis.

6.7 LLM-as-a-Judge

Zheng et al. (2023) showed that strong LLM judges can correlate substantially with human preferences in open-ended evaluation while also identifying biases such as position and verbosity. For scientific conclusions, relying on one LLM to judge another creates a risk of correlated errors.

6.8 Stability-Aware Prompt Optimization

Recent work treats prompt stability as a quality criterion. Chen et al. (2026) proposed semantic stability for evaluating auto-generated prompts and argued that prompt quality should account for response consistency rather than immediate task performance alone. This suggests a shift from optimizing prompts for accuracy alone toward accuracy, stability, and reliability.

7. Research Challenges and Limitations

7.1 Lack of a Standard Definition of Stability

Different studies use different notions of consistency, sensitivity, robustness, semantic similarity, and reproducibility. A standardized taxonomy should distinguish prompt sensitivity, decoding variability, contextual sensitivity, factual instability, inferential instability, and evaluation instability.

7.2 Semantic Equivalence Is Difficult to Establish

Two prompts can be linguistically different but semantically equivalent, while two superficially similar prompts can contain subtle differences in scope. Benchmark construction therefore requires human-validated semantic equivalence rather than automatic paraphrasing alone.

7.3 Evaluation Metrics May Create Artificial Sensitivity

Hua et al. (2025) provide an important counterpoint: some apparent prompt sensitivity may result from evaluation procedures themselves. Rigid answer matching and likelihood-based evaluation can exaggerate differences when responses are semantically equivalent.

7.4 Correctness Does Not Follow from Consistency

A model that repeatedly generates the same false conclusion is highly consistent but scientifically unreliable. TruthfulQA demonstrates that LLMs can reproduce false beliefs and misconceptions despite fluent responses (Lin et al., 2022).

7.5 Model Updates and Version Drift

Commercial and open models can change through updates, fine-tuning, safety adjustments, or inference changes. Scientific reproducibility therefore requires recording model name and version, date of inference, system prompt, user prompt, retrieved evidence, temperature and decoding settings, number of samples, output, and evaluation method.

7.6 Domain-Specific Risks

Prompt sensitivity may have different consequences across medicine, social science, engineering, and policy research. Generic prompt-stability benchmarks may therefore not adequately represent scientific risk.

8. Research Gaps and Future Directions

8.1 Scientific Conclusion Stability Benchmarks

There is a need for benchmarks specifically targeting research conclusions. A scientific benchmark could provide peer-reviewed papers, a research question, semantically equivalent prompts, expert-validated evidence summaries, expected conclusion categories, uncertainty labels, and expert judgments of acceptable inference.

8.2 A Conclusion Stability Index

Future work could develop a standardized Conclusion Stability Index (CSI) combining semantic agreement, evidential agreement, uncertainty agreement, and factual consistency. Such a metric would distinguish similar language from genuinely stable scientific inference.

8.3 Stability Under Evidence Perturbation

Prompt robustness should be studied together with evidence robustness by independently varying prompt, evidence, and sampling factors. This would help identify whether instability originates from wording, evidence selection, reasoning, decoding, or interaction effects.

8.4 Calibration of Scientific Uncertainty

Future LLM systems should be evaluated not only on whether the final conclusion is correct, but also on whether confidence changes appropriately when evidence changes. Strong evidence should lead to higher confidence, mixed evidence to moderate confidence, and limited evidence to lower confidence.

8.5 Adversarial Prompt Perturbation

Researchers should deliberately test synonyms, sentence restructuring, instruction order, positive versus neutral framing, negative versus neutral framing, role prompts, output formats, number of examples, and evidence-order changes.

8.6 Human-in-the-Loop Verification

Scientific use should remain human supervised for consequential conclusions. A robust workflow can require the LLM to provide a conclusion, supporting evidence, contradictory evidence, uncertainty, assumptions, limitations, and source references.

8.7 Provenance and Reproducibility

Every AI-assisted research result should ideally have an auditable provenance record: Research Question → Prompt → Evidence → Model → Output → Evaluation.

8.8 Beyond Prompt Engineering

Future scientific systems may combine LLMs with symbolic reasoning, statistical inference, retrieval systems, citation verification, formal hypothesis testing, structured databases, uncertainty estimation, and human expert review. The objective should be to move from prompt-dependent generation toward evidence-grounded scientific inference.

9. Conclusion

Prompt sensitivity represents a significant but underappreciated challenge for the use of LLMs in scientific research. Existing research demonstrates that relatively small changes in prompt wording, templates, demonstrations, and contextual organization can alter model behavior. At the same time, recent work cautions that some apparent sensitivity may arise from imperfect evaluation methodologies rather than genuine semantic instability.

For ordinary conversational applications, such variability may be relatively harmless. In scientific research, however, it can become consequential because LLMs are increasingly used to summarize evidence, evaluate hypotheses, identify research gaps, generate interpretations, and formulate conclusions. If semantically equivalent prompts yield materially different conclusions from the same evidence, then the resulting output cannot be considered fully robust.

The key distinction is between output similarity and conclusion stability. Scientific reliability requires more than consistent wording. It requires that factual claims, evidence interpretation, uncertainty, reasoning, and final inference remain reasonably stable under meaning-preserving perturbations.

Current approaches—including chain-of-thought prompting, self-consistency, few-shot examples, prompt ensembles, retrieval augmentation, structured evidence extraction, and stability-aware prompt optimization—provide useful mechanisms for reducing or diagnosing variability. However, none independently guarantees scientific correctness. Hallucination, evaluation artifacts, confirmation bias, model updates, and correlated errors remain important limitations.

Future research should establish standardized scientific benchmarks for prompt robustness, develop conclusion-level stability metrics, separate prompt sensitivity from sampling and context effects, and integrate factual verification with uncertainty calibration. The most reliable paradigm is unlikely to be one in which an LLM is simply given a carefully engineered prompt and trusted to produce a conclusion. Instead, LLM-assisted science should be designed as a reproducible, evidence-grounded, multi-stage process in which prompt robustness is explicitly measured and conclusions are independently verified.

Ultimately, prompt sensitivity should not be regarded merely as a prompt-engineering inconvenience. It is a methodological property of LLM-based scientific workflows. Treating it as such is essential if LLMs are to contribute to research without weakening the principles of reproducibility, transparency, evidential reasoning, and scientific reliability.

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